



UNIVERSITY OF
BIRMINGHAM

Category Theory

Midlands Graduate School 2026

Sergey Goncharov

April 13–17, 2026

Big Picture / Motivation

- **Category theory** is a “theory”, like **set theory**, like **group theory**, like **type theory**
- Types in functional programming $\leadsto$ sets (or set-like objects)
- Functions $\leadsto$ arrows/morphisms between types
- Category theory gives language to talk about **how** we compose computations, abstract patterns, and universal constructions that mirror programming constructs (pairs, function types, currying)
- It provides means of **abstraction** to reason and prove things **generically**, instead of on case-by-case basis

Course Outline

Part I: Sets and Warm-Up

Sets, functions, isomorphisms,
singletons, products, natural numbers

Part II: Categories & Functors

Definitions and examples, products,
coproducts, commutative diagrams,
universal properties

Part III: Natural Transformations

Natural transformations, exponentials,
Yoneda Theorem

Part IV: Adjunctions & Free Objects

Adjunctions, free objects

These slides:

www.sergey-goncharov.org/mgs-2026



Recommended Sources

- **Steve Awodey** — *Category Theory*, 2nd ed., Oxford University Press, 2010
 - [accessible introduction]
- **Emily Riehl** — *Category Theory in Context*, Dover, 2016
 - [freely available at <https://math.jhu.edu/~eriehl/161/context.pdf>]
- **Roy Crole** — *Categories for Types*, Cambridge University Press, 1993
 - [CS-oriented; Category theory for types & denotational semantics]
- **Saunders Mac Lane** — *Categories for the Working Mathematician*, 2nd ed., Springer, 1998
 - [timeless classics]

Sets and Warm-Up



Why Begin with Sets?

- Mathematics builds on sets
- Types in programming follow set intuition
- Category theory (standardly) also builds on sets
- But: what exactly is a set?
 - OK, that is too ambitious — what is it roughly?
- We start with sets to discover something deeper is hiding underneath

Naïve (Philosophical) Sets

- Set is collection of objects, like

$$X = \left\{ \text{⚽}, \text{🐕}, \pi, \mathbb{N}, \text{♥} \right\}$$

- We write $x \in X$ to say that x is **member** of X , e.g. $\pi \in X$
- **Empty set** $\emptyset = \{ \}$ is a set of no elements
- There are various mechanisms to construct larger sets
- **Set comprehension**

$$\{x \in S \mid P(x)\} \quad \text{e.g.} \quad \text{evens} = \{n \in \mathbb{N} \mid \exists m \in \mathbb{N}. n = 2m\}$$

Intuition: Sets are containers

Finite Sets v.s. Finite Lists

- **(Finite) lists** over set X : $\text{List}(X) = \{[x_1, \dots, x_n] \mid n \in \mathbb{N}, x_1 \in X, \dots, x_n \in X\}$
- **Notation:** $[]$ — **empty list**, $x :: xs$ — list with **head** x and **tail** xs . E.g. $x :: [] = [x]$
- For contrast/analogy: Finite sets are lists, where we additionally enable
 - **Commutativity:** $[\dots, x, y, \dots] \equiv [\dots, y, x, \dots]$
 - **Idempotence:** $[\dots, x, x, \dots] \equiv [\dots, x, \dots]$
- **(Finite) Multisets** = lists with commutativity, but not idempotence. Thus

Lists < Multisets < Sets

Sets (Slightly More) Formally

- Set theory postulates: everything is a set
- Only primitive relation between sets is ' $\in$ '

We then derive:

- **Natural numbers:** $0 = \emptyset$, $1 = \{0, \emptyset\} = \{\emptyset\}$, $2 = \{1, \emptyset\} = \{\{\emptyset\}, \emptyset\}, \dots$
- Set inclusion: $X \subseteq Y$ if $\forall x \in X. x \in Y$
 - **Extensionality axiom:** $X = Y$ if and only if $X \subseteq Y$ and $Y \subseteq X$
- **Powerset:** $Y \in \mathcal{P}(X)$ if and only if $Y \subseteq X$
- **Relations:** $R \subseteq X \times Y$
- **Functions:** $f: X \rightarrow Y$ — such relations $R \subseteq X \times Y$ that for every $x \in X$ there is unique $y \in Y$ with $x R y$
 - So, in set theory: Sets $\leadsto$ Relations $\leadsto$ Functions

Empty Set and Singleton Sets

- No controversy about empty set: $\emptyset = \{ \}$ — set with no elements
- But what is one-element set? $\{\oplus\}$? $\{\text{🐕}\}$? $\{\pi\}$?
- $\{\emptyset\}$ is good candidate
- For every set S , $\{S\}$ is also singleton set
- Singleton sets are characterized by property: they are sets with exactly one element
- $\{\emptyset\}$ is a concrete **representative** in the **class** of all singletons

Note on Classes

- All sets do not form a set, but a **class**

$\text{Set} < \text{Class}$

- Division into sets and proper classes is way to avoid paradoxes
- Other approach: **universes**

$\text{Set}_0 < \text{Set}_1 < \text{Set}_2 < \dots$

where the union of all sets in Set_i is in Set_{i+1}

- By saying “set” we will imagine set from Set_i , and by saying “class” we will imagine set from Set_{i+1}

Natural Numbers as Sets

- In computer science, we count from 0: $\mathbb{N} = \{0, 1, \dots\}$!
- **Von Neumann numerals:** $0 := \emptyset, n + 1 := n \cup \{n\}$ So:

$$0 = \emptyset, \quad 1 = \{\emptyset\}, \quad 2 = \{\emptyset, \{\emptyset\}\}, \quad 3 = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}, \dots$$

This is very convenient, because $n + 1 = \{0, \dots, n\}$

- Thus, we can — and we will! — identify natural numbers with sets
- Alternatively **Zermelo numerals:** $0 := \emptyset, n + 1 := \{n\}$. This is less convenient

Ordered Pairs and Products

We would like to define **Cartesian products** $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$. But what is (x, y) ?

- **Kuratowski definition:** $(x, y) := \{\{x\}, \{x, y\}\}$
- Then (**PROVE IT**):

$$(x, y) = (x', y') \iff x = x' \text{ and } y = y'.$$

- Using this, we can define:

$$X \times Y := \left\{ \{\{x\}, \{x, y\}\} \subseteq \mathcal{P}(X \cup Y) \mid x \in X, y \in Y \right\}$$

Pairs and Cartesian products are **constructed objects**!

Disjoint Unions

- **Union of sets:** $x \in X \cup Y$ iff $x \in X$ or $x \in Y$
- How to define **Disjoint Union** $X + Y (= X \uplus Y)$?
- Maybe $X + Y = \{(x, \text{🔑}) \mid x \in X\} \cup \{(x, \text{🔧}) \mid y \in Y\}$?

Disjoint Unions

- **Union of sets:** $x \in X \cup Y$ iff $x \in X$ or $x \in Y$
- How to define **Disjoint Union** $X + Y (= X \uplus Y)$?
- Maybe $X + Y = \{(x, \text{🔑}) \mid x \in X\} \cup \{(x, \text{🔧}) \mid y \in Y\}$?
- Maybe $X + Y = \{(x, \text{🍔}) \mid x \in X\} \cup \{(x, \text{🐘}) \mid y \in Y\}$?

Disjoint Unions

- **Union of sets:** $x \in X \cup Y$ iff $x \in X$ or $x \in Y$
- How to define **Disjoint Union** $X + Y (= X \uplus Y)$?
- Maybe $X + Y = \{(x, \text{🔑}) \mid x \in X\} \cup \{(x, \text{🔧}) \mid y \in Y\}$?
- Maybe $X + Y = \{(x, \text{🍔}) \mid x \in X\} \cup \{(x, \text{🐘}) \mid y \in Y\}$?
- More seriously: $X + Y = \{(x, 0) \mid x \in X\} \cup \{(x, 1) \mid y \in Y\}$

Now compute with von Neumann numerals: $1 = \{0\}$, $2 = \{0, 1\}$. Then

$$1 + 1 = \{(1, 0), (1, 1)\} = \{\{ \{1\}, \{1, 0\} \}, \{ \{1\} \} \}$$

This is **not** equal to $2 = \{0, 1\}$! So

$$1 + 1 \neq 2$$

Functions between Sets

- Function $f: X \rightarrow Y$, assigns each $x \in X$ a unique $f(x) \in Y$
- Functions $f: X \rightarrow Y$ are special relations $f \subseteq X \times Y$
- Identity function $id_X: X \rightarrow X$, $id_X(x) = x$
- Equality of functions $f = g$ if for all x , $f(x) = g(x)$
 - This is provable principle (**PROVE IT**), called **function extensionality**

Composition:

$$(g \circ f)(x) = g(f(x))$$

Proposition: $f \circ id = f$, $id \circ f = f$ and $f \circ (g \circ h) = (f \circ g) \circ h$ (associativity)

Bijection vs Isomorphism

Function $f: X \rightarrow Y$ is

- **Injection** if $f(x) = f(x')$ entails $x = x'$
- **Surjection** if for every $y \in Y$ there is $x \in X$, such that $f(x) = y$
- **Bijection** if it is both injection and surjection

Alternative view: $f: X \rightarrow Y$ is **isomorphism** if for some $f^{-1}: Y \rightarrow X$,

$$f \circ f^{-1} = id_Y \quad \text{and} \quad f^{-1} \circ f = id_X$$

Proposition: f is bijection if and only if f is isomorphism

Some Properties of Isomorphisms

It follows that (**PROVE IT**)

- f^{-1} is unique if exists
- If f is isomorphism, so is f^{-1}
- id is isomorphism and $id^{-1} = id$
- Composition of isomorphisms $f \circ g$ is isomorphism and $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

These are derivable from definition, without using that isomorphism = bijection!

Notation: $X \cong Y$ if there is isomorphism $f: X \rightarrow Y$

Up-to Isomorphism

- Big innovation of category theory: work with things via their descriptions
- We narrow down objects of interest as those that “fit best” to such description
- Such descriptions together with their best-fit condition are called **universal properties**
- They determine objects not uniquely, but **up-to isomorphism**

Example. A **singleton set** 1 is characterised by: there is exactly one function $X \rightarrow 1$ for every set X

- $\{\emptyset\}$, $\{\{\emptyset\}\}$, $\{\odot\}$ all satisfy this — and are all isomorphic
- Any two singletons are isomorphic, so we speak of **the singleton up to isomorphism**

Induction/Recursion for Natural Numbers

- Set theory postulates existence of natural numbers $\mathbb{N}$ as smallest set that contains $0 = \emptyset$ and closed under $n \mapsto n + 1 := n \cup \{n\}$
- **Induction** is paraphrasing this: if $P \subseteq \mathbb{N}$ with $0 \in P$ and $n \in P \Rightarrow n+1 \in P$, then $P = \mathbb{N}$
 - Because predicates over $\mathbb{N} \iff$ subsets of $\mathbb{N}$:

$$P(0) \wedge (\forall n. P(n) \Rightarrow P(n+1)) \Rightarrow \forall n. P(n)$$

- **(Structural) Recursion**: for any $a \in A$ and $h: A \rightarrow A$, there is unique $f: \mathbb{N} \rightarrow A$ with

$$f(0) = a \qquad f(n+1) = h(f(n))$$

- **Primitive recursion** is derivable from recursion (**PROVE IT**): for any $g: X \rightarrow A$ and $h: A \times X \times \mathbb{N} \rightarrow A$, there is unique $f: X \times \mathbb{N} \rightarrow A$ with

$$f(x, 0) = g(x) \qquad f(x, n+1) = h(f(x, n), x, n)$$

Induction/Recursion for Lists

- Given set X , $\text{List}(X)$ is constructable as smallest set that contains $[]$ and closed under $xs \mapsto x :: xs$ for $x \in X$
- **Induction:** $P([]) \wedge (\forall x \in X, xs. P(xs) \Rightarrow P(x :: xs)) \Rightarrow \forall xs. P(xs)$
- **(Structural) Recursion:** for any $a \in A$ and $h: X \times A \rightarrow A$, there is unique $f: \text{List}(X) \rightarrow A$ with

$$f([]) = a \qquad f(x :: xs) = h(x, f(xs))$$

Questions:

- What is primitive recursion for lists?
- Why it is derivable (**PROVE IT**)
- Define *length* and concatenation recursively

Remarks:

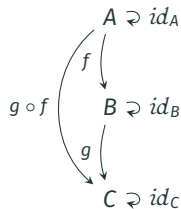
- Structural recursion $\neq$ general recursion (no arbitrary self-calls)
- Induction — proof principle; recursion — definition principle

Categories & Functors

Definition of Category

Definition: **Category** $\mathcal{C}$ consists of:

- **Class of objects** $\text{Ob}(\mathcal{C})$
- For every A, B **set of morphisms** $\text{Hom}_{\mathcal{C}}(A, B)$. Write $f: A \rightarrow B$
- **Composition**: if $f: A \rightarrow B$ and $g: B \rightarrow C$ then $g \circ f: A \rightarrow C$
- **Identity morphisms** $\text{id}_A: A \rightarrow A$



Subject to **associativity** $(h \circ g) \circ f = h \circ (g \circ f)$ and **identity laws** $\text{id} \circ f = f = f \circ \text{id}$

- We can imagine categories as (multi)graphs: objects = nodes, morphisms = edges
- We defined **locally small categories**, but see Slide 9 for comment on size issues!

Sets as Category

Category of sets **Set**

- **Objects:** sets
- **Morphisms:** functions between sets
- Composition: usual function composition
- Identities: identity functions

Further set-like categories (objects = sets, identities = identity functions):

- Relations (**Rel**) — morphisms are relations
- Partial functions
- Injections
- Isomorphisms
- ...

Category Pos

- **Objects:** partially ordered sets $(P, \leq)$
- **Morphisms:** monotone functions

$$f: (P, \leq_P) \rightarrow (Q, \leq_Q)$$

such that

$$x \leq_P y \Rightarrow f(x) \leq_Q f(y)$$

Example:



Category Mon

- **Objects:** monoids $(M, \cdot, e)$
- **Morphisms:** **monoid homomorphisms** are maps $f: M \rightarrow N$, such that unit and multiplication are preserved:

$$f(x \cdot y) = f(x) \cdot f(y) \qquad f(e_M) = e_N$$

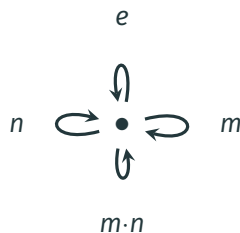
Example: $length: (\text{List}(A), ++, []) \rightarrow (\mathbb{N}, +, 0)$ — preserves unit and multiplication:

$$\begin{aligned} length([]) &= 0 \\ length(xs ++ ys) &= length(xs) + length(ys) \end{aligned}$$

Monoid as One-Object Category

Fix some monoid $(M, \cdot, e)$. We can turn it into category!

- **Objects:** — exactly one, say $\bullet$
- **Morphisms:** $\text{Hom}(\bullet, \bullet) = M$
- Composition is given by monoid multiplication
- Identity given by e



This is equivalence: Monoids $\iff$ one-object categories (**PROVE IT**)

Monoid homomorphisms $\iff$ functors between one-object categories (**PROVE IT**)

Posets as Categories

Fix poset $(P, \leq)$. It is also category:

- **Objects:** elements of P
- **Morphisms:** $\text{Hom}(x, y) = \{\star\}$ if $x \leq y$ and $\text{Hom}(x, y) = \emptyset$ otherwise
(at most one morphism between two objects)
- Unit of composition = reflexivity
- Associativity of composition = transitivity

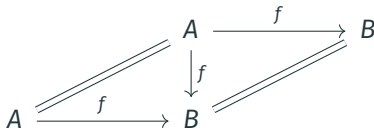
Hint: Categorical notions become **order-theoretic** when instantiated to poset-categories.
To understand a complex categorical notion, instantiate it to poset-category!

Example Categories: Summary

Category	Objects	Morphisms	Composition
Set	sets	functions	function composition
Pos	posets	monotone maps	function composition
Mon	monoids	homomorphisms	function composition
Rel	sets	relations $R \subseteq X \times Y$	relational composition
$(P, \leq)$	elements of P	$\star$ if $x \leq y$	transitivity
$(M, \cdot, e)$	$\{\star\}$	elements of M	monoid multiplication

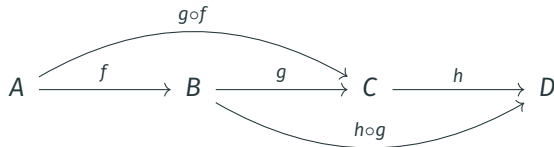
Commutative Diagrams

- For $f: A \rightarrow B$, we can express $f \circ id_A = f$ and $id_B \circ f = f$ diagrammatically:



(‘ $=$ ’ denotes ‘ $\xrightarrow{id}$ ’)

- Associativity:



But that is redundant, because we read sequence f, g as $g \circ f$ — associativity is baked in!

Isomorphisms

Morphism $f: A \rightarrow B$ is an **isomorphism** if there exists $f^{-1}: B \rightarrow A$ with $f^{-1} \circ f = id_A$ and $f \circ f^{-1} = id_B$:



- A and B are **isomorphic**, written $A \cong B$, if such f exists
- f^{-1} is unique: if $g \circ f = id_A$ and $f \circ g = id_B$, then

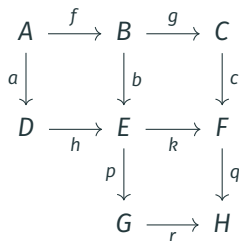
$$g = g \circ id_B = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = id_A \circ f^{-1} = f^{-1}$$

- In **Set**: isomorphism = bijection; in **Pos**: order-isomorphism; in **Mon**: monoid isomorphism
- **Group** is one-object category in which every morphism is isomorphism (**PROVE IT**)

Commutative Diagrams and Equational Reasoning

A diagram **commutes** if every directed path with the same start and end gives the same composite.

If all three squares commute:



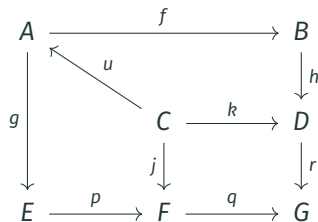
$$b \circ f = h \circ a, \quad c \circ g = k \circ b, \quad q \circ k = r \circ p$$

then the outer paths from A to H agree:

$$\begin{aligned} q \circ c \circ g \circ f &= q \circ (k \circ b) \circ f \\ &= (q \circ k) \circ (b \circ f) \\ &= (r \circ p) \circ (h \circ a) \\ &= r \circ p \circ h \circ a \end{aligned}$$

Thus, commutative diagrams = pictorial representations of equational proofs!

Not Every Diagram Proves Something



Commuting cells (by assumption):

$$h \circ f \circ u = k$$

$$p \circ g \circ u = j$$

$$r \circ k = q \circ j$$

Chaining all three:

$$(r \circ h \circ f) \circ u = r \circ k$$

$$= q \circ j$$

$$= (q \circ p \circ g) \circ u$$

But this does not imply $r \circ h \circ f = q \circ p \circ g$, because we generally cannot discard u !

Terminal and Initial Objects

In category $\mathcal{C}$:

- **Terminal object** 1: for every object A there is exactly one morphism $A \rightarrow 1$
- **Initial object** 0: for every object A there is exactly one morphism $0 \rightarrow A$

Examples:

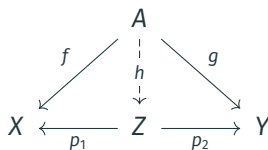
- In **Set**: any singleton $\{*\}$ is terminal; $\emptyset$ is initial
- In **Pos**: any one-element poset is terminal; $\emptyset$ is initial
- In **Mon**: trivial monoid $\{e\}$ is terminal and initial
- In every poset category: terminal = **greatest element**; initial = **least element**

Key viewpoint: elements of X in **Set** correspond to arrows $1 \rightarrow X$, i.e.

$\text{elements}(X) \cong \text{Hom}(1, X)$. So, $f(x)$ must be interpreted as composition $1 \xrightarrow{x} X \xrightarrow{f} Y$!

Product Universal Property

- **Object:** a product of X and Y is an object Z
- **Diagram (span):** morphisms $p_1: Z \rightarrow X$ and $p_2: Z \rightarrow Y$
- **Universal property:** for any $f: A \rightarrow X$ and $g: A \rightarrow Y$ there is a **unique** $h: A \rightarrow Z$ with $p_1 \circ h = f$ and $p_2 \circ h = g$:



Selected product $X \times Y$: specific **choice** of product for every pair (X, Y) ; all products of X and Y are **uniquely isomorphic**

Notation: With $X \times Y$, use fst , snd for the chosen projections; $\langle f, g \rangle$ for the induced h

Examples of Products

- **Set:** $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$; projections $(x, y) \mapsto x$ and $(x, y) \mapsto y$; pairing $\langle f, g \rangle(x) = (f(x), g(x))$
- **Pos:** $X \times Y$ with pointwise order: $(x, y) \leq (x', y')$ iff $x \leq x'$ and $y \leq y'$
- **Mon:** componentwise monoid $(M \times N, \cdot, (e_M, e_N))$ where

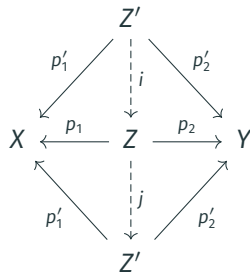
$$(m, n) \cdot (m', n') = (m \cdot m', n \cdot n')$$

- **Poset category:** product of x and y = **meet** $x \wedge y$ (=greatest lower bound), when it exists

Products are Up-To-Isomorphism

Let (Z, p_1, p_2) and (Z', p'_1, p'_2) both be products of X and Y :

- Universal property of Z : $\exists! i: Z' \rightarrow Z$ with $p_1 \circ i = p'_1$,
 $p_2 \circ i = p'_2$
- Universal property of Z' on $\text{span}(p_1, p_2)$: $\exists! j: Z \rightarrow Z'$
with $p'_1 \circ j = p_1$, $p'_2 \circ j = p_2$
- By uniqueness: $j \circ i = id_{Z'}$
- By symmetric argument: $i \circ j = id_Z$, so $Z \cong Z'$



Example: Take selected product $(X \times Y, fst, snd)$. Then $(Y \times X, snd, fst)$ is also product of X and Y — isomorphism $\text{swap} = \langle snd, fst \rangle: X \times Y \xrightarrow{\sim} Y \times X$

Axioms of Binary Products

For selected products, one can show (**PROVE IT**) :

$$(\beta_1) \text{ } fst \circ \langle f, g \rangle = f$$

$$(\beta_2) \text{ } snd \circ \langle f, g \rangle = g$$

$$(\eta) \text{ } \langle fst, snd \rangle = id_{X \times Y}$$

$$(\text{nat}) \text{ } \langle f, g \rangle \circ h = \langle f \circ h, g \circ h \rangle$$

Theorem: This is a **sound** and **complete** axiomatization of binary products (**PROVE IT**)

Example (Uniqueness of pairing): if $fst \circ k = f$ and $snd \circ k = g$, then

$$k = id \circ k \stackrel{(\eta)}{=} \langle fst, snd \rangle \circ k \stackrel{(\text{nat})}{=} \langle fst \circ k, snd \circ k \rangle \stackrel{(\text{assm.})}{=} \langle f, g \rangle$$

Dual Category

Given category $\mathcal{C}$, the **opposite category** $\mathcal{C}^{\text{op}}$ has:

- Same objects as $\mathcal{C}$
- Morphisms reversed: $f: A \rightarrow B$ in $\mathcal{C}$ becomes $f^{\text{op}}: B \rightarrow A$ in $\mathcal{C}^{\text{op}}$
- Composition: $g^{\text{op}} \circ f^{\text{op}} = (f \circ g)^{\text{op}}$

Principle of Duality: any theorem proved in $\mathcal{C}$ yields a dual theorem in $\mathcal{C}^{\text{op}}$ — just reverse all arrows

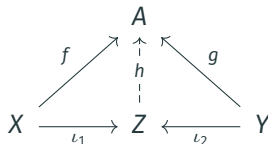
Examples:

- Poset $(P, \leq)^{\text{op}} = \text{poset with reversed order } (P, \geq)$
- Initial objects in $\mathcal{C}$ = terminal objects in $\mathcal{C}^{\text{op}}$

Coproduct via Duality

Coproduct = product in $\mathcal{C}^{\text{op}}$ — obtained by reversing all arrows in the product definition

- **Object:** a coproduct of X and Y is an object Z
- **Diagram (cospan):** morphisms $\iota_1: X \rightarrow Z$ and $\iota_2: Y \rightarrow Z$
- **Universal property:** for any $f: X \rightarrow A$ and $g: Y \rightarrow A$ there is a **unique** $h: Z \rightarrow A$ with $h \circ \iota_1 = f$ and $h \circ \iota_2 = g$:



Selected coproduct $X + Y$: choice of coproduct for every pair; notation inl, inr for chosen injections; $[f, g]$ — **co-pairing** — for the induced h

Example: In **Set**, $X + Y =$ disjoint union (**PROVE IT**)

Axioms of Binary Coproducts

For selected coproducts, one can show:

$$(\beta_1) [f, g] \circ \text{inl} = f$$

$$(\beta_2) [f, g] \circ \text{inr} = g$$

$$(\eta) [\text{inl}, \text{inr}] = \text{id}_{X+Y}$$

$$(\text{nat}) h \circ [f, g] = [h \circ f, h \circ g]$$

Theorem: This is **sound** and **complete** axiomatization of binary coproducts

Proof: Duality

Example: Every $f: X + Y \rightarrow Z$ has form $[f_1, f_2]$ for $f_1: X \rightarrow Z, f_2: Y \rightarrow Z$. Indeed:

$$f = f \circ \text{id} \stackrel{(\eta)}{=} f \circ [\text{inl}, \text{inr}] \stackrel{(\text{nat})}{=} [f \circ \text{inl}, f \circ \text{inr}]$$

Categories & Functors

Functors

Definition of Functor

(Covariant) Functor $F: \mathcal{C} \rightarrow \mathcal{D}$ acts on objects and morphisms as follows:

$$A \in \text{Ob}(\mathcal{C}) \mapsto F(A) \in \text{Ob}(\mathcal{D}) \qquad \begin{array}{ccc} A & & F(A) \\ f \downarrow & \longmapsto & \downarrow F(f) \\ B & & F(B) \end{array}$$

preserving composition and identities:

$$F(id_A) = id_{F(A)} \qquad F(g \circ f) = F(g) \circ F(f)$$

- Functor $F: \mathcal{C} \rightarrow \mathcal{C}$ is called **endofunctor**
- Functors compose: $(GF)(A) = G(F(A))$, $(GF)(f) = G(F(f))$; categories and functors form **Cat** — **category of categories**, modulo size issues

Examples of Functors

Endofunctors on Set:

- Powerset: $\mathcal{P}(X)$; on $f: X \rightarrow Y$ acts as direct image $\mathcal{P}(f)(S) = \{f(x) \mid x \in S\}$
- List: $\text{List}(X)$ = finite lists over X ; $\text{List}(f)([x_1, \dots, x_n]) = [f(x_1), \dots, f(x_n)]$
- Maybe: $X \mapsto X + \{\text{Nothing}\}$:
 - $\text{Maybe}(f)(\text{inl } x) = \text{inl}(f(x))$
 - $\text{Maybe}(f)(\text{inr } x) = \text{inr } x$

Between different categories:

- **Forgetful functor** $G: \mathbf{Mon} \rightarrow \mathbf{Set}$: sends $(M, \cdot, e)$ to M , homomorphisms to functions
- **Free monoid functor** $F: \mathbf{Set} \rightarrow \mathbf{Mon}$: sends X to $\text{List}(X)$

Bifunctors

Bifunctor $F: \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{E}$: functor on product category. Unraveling definition:

- $F(id_A, id_B) = id_{F(A,B)}$
- $F(f' \circ f, g' \circ g) = F(f', g') \circ F(f, g)$

Example: Binary product $(- \times -): \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ (if $\mathcal{C}$ has binary products): on morphisms $f: A \rightarrow A', g: B \rightarrow B'$, gives

$$f \times g = \langle f \circ fst, g \circ snd \rangle: A \times B \rightarrow A' \times B'$$

Contravariant and Mixed-Variance Functors

Contravariant functor $F: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$: sends $f: A \rightarrow B$ to $F(f): F(B) \rightarrow F(A)$; reverses composition: $F(g \circ f) = F(f) \circ F(g)$

Mixed variance $F: \mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \mathcal{E}$: **contravariant** in first argument, **covariant** in second

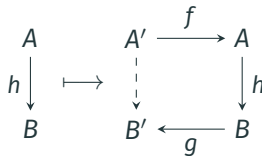
Example: $\text{Hom}(-, -): \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Set}$ is mixed-variance (see next slide)

Duality: contravariant functor $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ is the same as covariant functor $\mathcal{C} \rightarrow \mathcal{D}^{\text{op}}$; choice of perspective is a matter of convention

Hom-Functor

Hom-functor $\text{Hom}(-, -): \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Set}$: For given $f: A' \rightarrow A$ and $g: B \rightarrow B'$, $\text{Hom}(f, g)$ sandwiches input between f and g :

$$\text{Hom}(f, g): \text{Hom}(A, B) \rightarrow \text{Hom}(A', B')$$



This specializes in two ways:

- $\text{Hom}(A, -): \mathcal{C} \rightarrow \mathbf{Set}$ — covariant; acts by **post**composition: $h \mapsto f \circ h$
- $\text{Hom}(-, B): \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ — contravariant; acts by **pre**composition: $h \mapsto h \circ f$

Contravariant powerset functor sends $X \mapsto \mathcal{P}(X)$ and $f: X \rightarrow Y$ to $f^{-1}[-]: \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$

This is isomorphic to $\text{Hom}_{\mathbf{Set}}(X, 2)$ (**PROVE IT**)

Full and Faithful Functors

Functor $F: \mathcal{C} \rightarrow \mathcal{D}$ induces maps on hom-sets:

$$F_{A,B}: \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(FA, FB), \quad f \mapsto F(f)$$

- F is **faithful** if every $F_{A,B}$ is injective
- F is **full** if every $F_{A,B}$ is surjective
- F is **fully faithful** if every $F_{A,B}$ is a bijection

Examples:

- Forgetful $G: \mathbf{Mon} \rightarrow \mathbf{Set}$: **faithful** (distinct homomorphisms remain distinct) but not full (not every function is a homomorphism)
- Inclusion $(P, \leq) \hookrightarrow \mathbf{Set}$ via poset-as-category: **faithful** (Hom has at most one element), not full in general

Natural Transformations



Definition: Natural Transformation

For functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$, **natural transformation** α assigns to each object X morphism $\alpha_X: F(X) \rightarrow G(X)$ (**component** at X) such that for every $f: X \rightarrow Y$ **naturality square** commutes:

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \alpha_X \downarrow & & \downarrow \alpha_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

$$G(f) \circ \alpha_X = \alpha_Y \circ F(f)$$

- Write $\alpha: F \Rightarrow G$
- α is a **natural isomorphism** if each α_X is an isomorphism; write $F \cong G$

Programming intuition: Naturality $\approx$ **parametric polymorphism** — a parametrically polymorphic function $\forall A. F(A) \rightarrow G(A)$ is automatically natural

Examples: Natural Transformations

Natural Transformations between Endofunctors on Set:

- **Identity:** $id: Id \Rightarrow Id$ — trivially natural
- **Singleton:** $sngl: Id \Rightarrow List$, $sngl_X(x) = [x]$:

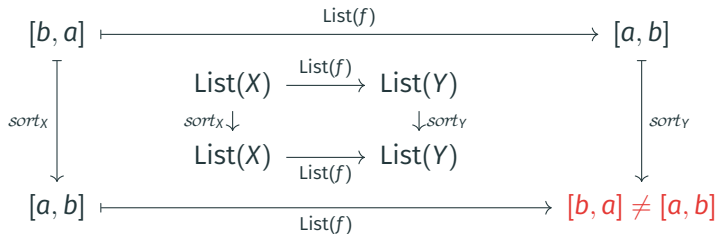
$$\begin{array}{ccccc} x \in X & \xrightarrow{\quad f \quad} & & & f(x) \\ \downarrow sngl_X & & X \xrightarrow{f} Y & & \downarrow sngl_Y \\ & & sngl_X \downarrow \quad \downarrow sngl_Y & & \\ & & List(X) \xrightarrow{List(f)} List(Y) & & \\ & & \downarrow sngl_Y & & \\ [x] & \xrightarrow{List(f)} & & & [f(x)] \end{array}$$

- **Flatten:** $\mu: List\ List \Rightarrow List$, $\mu_X([[a, b], [c]]) = [a, b, c]$ — natural (**PROVE IT**)
- **Reverse:** $rev: List \Rightarrow List$, $rev_X([x_1, \dots, x_n]) = [x_n, \dots, x_1]$ — natural (**PROVE IT**)

Non-Example: Sort is Not Natural

Let $sort_X$ sort $List(X)$ by some total order on X

Counterexample: Let $f: \{a, b\} \rightarrow \{a, b\}$ swap a and b where $a < b$:



Intuitively, components of natural transformations must be agnostic about properties of objects not preserved by morphisms

Functor Category $[\mathcal{C}, \mathcal{D}]$

- **Objects:** functors $\mathcal{C} \rightarrow \mathcal{D}$
- **Morphisms:** natural transformations $F \Rightarrow G$
- **Composition:** for $\alpha: F \Rightarrow G$ and $\beta: G \Rightarrow H$, define $(\beta \circ \alpha)_X = \beta_X \circ \alpha_X$

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \alpha_X \downarrow & & \downarrow \alpha_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \\ \beta_X \downarrow & & \downarrow \beta_Y \\ H(X) & \xrightarrow{H(f)} & H(Y) \end{array}$$

- **Identity:** $(id_X: X \rightarrow X)_X$

Natural Isomorphisms

$\alpha: F \Rightarrow G$ is **natural isomorphism** if each component α_X is isomorphism

- Equivalently: α is isomorphism in $[F, G]$ (**PROVE IT**)
- F and G are **naturally isomorphic**, written $F \cong G$, if such α exists

Examples (**PROVE IT**):

- Swap: $(- \times -) \cong (- \times -)$ via $swap_{X,Y}: X \times Y \rightarrow Y \times X$, $swap_{X,Y} = \langle snd, fst \rangle$
- Products are associative: $(X \times Y) \times Z \cong X \times (Y \times Z)$ naturally in X, Y, Z
- In **Set**: $\text{Hom}(1, X) \cong X$ naturally in X (elements as arrows)

Practical Remarks

- Often, instead “ α is natural transformation $F \Rightarrow G$ ” one says “family $\alpha_X: F(X) \rightarrow G(X)$ is **natural in X** ”
- Same for bi-functors: $\alpha_{X,Y}: F(X, Y) \rightarrow G(X, Y)$ is natural in X and Y , etc
- Naturality in $(X, Y) \Leftrightarrow$ naturality in X + naturality in Y (**PROVE IT**):

$$\begin{array}{ccc} F(X, Y) & \xrightarrow{\alpha_{X,Y}} & G(X, Y) \\ F(f, Y) \downarrow & & \downarrow G(f, Y) \\ F(X', Y) & \xrightarrow{\alpha_{X',Y}} & G(X', Y) \end{array}$$

$$\begin{array}{ccc} F(X, Y) & \xrightarrow{\alpha_{X,Y}} & G(X, Y) \\ F(X, g) \downarrow & & \downarrow G(X, g) \\ F(X, Y') & \xrightarrow{\alpha_{X,Y'}} & G(X, Y') \end{array}$$

Further Remarks

Composition of natural transformations and functors (**whiskering**):

- $K\alpha: (K\alpha)_X = K(\alpha_X)$ (post-whiskering)
- $\beta H: (\beta H)_X = \beta_{H(X)}$ (pre-whiskering)

Despite symmetry: $K\alpha$ is α wrapped in K , while βH — alternatively β_H — only varies component of β . So, more suggestively:

$$(K\alpha)_X = K(\alpha_X) \quad (\beta_H)_X = \beta_{HX}$$

Often, natural transformations are conflated with their components, e.g:

$$\begin{array}{ccc} H(F(X)) & \xrightarrow{\beta} & K(F(X)) \\ & \searrow H\alpha & \swarrow K\alpha \\ & H(G(X)) & \xrightarrow{\beta} K(G(X)) \end{array}$$

Natural Transformations



Exponentials

Hom's v.s. Exponentials

- **External Hom**: $\text{Hom}_{\mathcal{C}}(X, Y)$ lives outside $\mathcal{C}$ — $\text{Hom}: \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Set}$
- **Internal Hom** = **Exponential object** Y^X — representing same morphisms inside $\mathcal{C}$:

$$(-)^{(-)}: \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathcal{C}$$

- **Example in Pos**:
 - $\text{Hom}_{\mathbf{Pos}}(X, Y)$ = monotone maps $X \rightarrow Y$ — set, outside **Pos**
 - Y^X in **Pos** = same maps with **pointwise order**: $f \leq g \Leftrightarrow \forall x. f(x) \leq g(x)$ — inside **Pos**

Exponential Object: Definition

Fix object X in a category $\mathcal{C}$ with binary products

- **Object:** Y^X (also written $[X, Y]$ or $X \Rightarrow Y$)
- **Arrows:** evaluation morphism $ev_{X,Y}: Y^X \times X \rightarrow Y$
- **Universal property:** for any $f: Z \times X \rightarrow Y$, there exists a **unique** $curry_{X,Y,Z}(f): Z \rightarrow Y^X$ such that:

$$\begin{array}{ccc} Z \times X & \xrightarrow{curry_{X,Y,Z}(f) \times id_X} & Y^X \times X \\ & \searrow f & \downarrow ev_{X,Y} \\ & & Y \end{array}$$

Same caveat as before about “an exponential” and “selected exponential” applies

Axioms of Exponentials

For selected exponentials, one can show (**PROVE IT**) :

$$(\beta) \quad ev \circ (curry(f) \times id) = f$$

$$(\eta) \quad curry(ev) = id$$

$$(\text{nat}) \quad curry(f) \circ g = curry(f \circ (g \times id))$$

Theorem: This axiomatization is sound and complete(**PROVE IT**)

Exponential Object: Basic Properties

- For $g: Z \rightarrow Y^X$, define $uncurry(g)$ as

$$Z \times X \xrightarrow{g \times id_X} Y^X \times X \xrightarrow{ev_{X,Y}} Y$$

- Then, $uncurry = curry^{-1}$, witnessing isomorphism (**PROVE IT**)

$$\text{Hom}(Z \times X, Y) \cong \text{Hom}(Z, Y^X) :$$

- $curry, uncurry$ are natural in all involved parameters, e.g. (nat) is precisely naturality of $curry_{X,Y,Z}$ in Z :

$$\begin{array}{ccc} \text{Hom}(Z \times X, Y) & \xrightarrow{curry} & \text{Hom}(Z, Y^X) \\ \downarrow - \circ (h \times id_X) & & \downarrow - \circ h \\ \text{Hom}(Z' \times X, Y) & \xrightarrow{curry} & \text{Hom}(Z', Y^X) \end{array}$$

Examples of Exponentials

- **Set:** Y^X = set of functions $X \rightarrow Y$; $ev(g, x) = g(x)$; $curry(f)(z)(x) = f(z, x)$
- **Pos:** Y^X = only **monotone functions** $X \rightarrow Y$ (!) with pointwise order:

$$g \leq h \Leftrightarrow \forall x. g(x) \leq h(x)$$

curry and *ev* are defined by same expressions

- **Poset categories:** recall, $x \rightarrow y$ iff $x \leq y$
 - exponential $y^x = (x \Rightarrow y)$ is the **Heyting implication**: largest z with $z \wedge x \leq y$
 - *ev* is **modus ponens** $z \wedge (z \Rightarrow y) \leq y$
- **Non-Example:** in **Mon**, exponentials generally do not exist (**PROVE IT**)

Cartesian Closed Categories

Category $\mathcal{C}$ is **Cartesian closed** (CCC) if it has:

- terminal object 1
- binary products $X \times Y$ for all X, Y
- exponentials Y^X for all X, Y

CCC are exactly categorical models of **typed λ -calculus** (with product types)!

Examples:

- **Set** — suitable for semantics of higher-order without divergence (cf. Paul's course)
- Groupoids — like sets, but proof-relevant (cf. Thorsten's course)
- Categories of domains (e.g. **Scott domains**) for modeling divergence

Natural Transformations

Yoneda Theorem (Lemma)

Polynomial Functors

Polynomial functors are functors of the form:

$$P(X) = \coprod_{i \in I} X^{n_i} \quad n_i \in \mathbb{N}$$

where $X^n = \underbrace{X \times \cdots \times X}_n$ (**powers**) and $\coprod_{i \in I}$ are iterated products and coproducts

Example: Polynomial functors model **algebraic signatures**. For example, monoid $(M, e, \cdot)$ (in **Set**) is represented by morphism (function)

$$f: P(M) \rightarrow M \quad \text{where} \quad P(X) = 1 + X \times X$$

Recall, necessarily $f = [f_1, f_2]$ with $f_1: 1 \rightarrow M$, $f_2: M \times M \rightarrow M$. $f_1(\star) = e$, $f_2(x, y) = x \cdot y$

Example: List $X = 1 + X + X^2 + \dots = \coprod_{n \in \mathbb{N}} X^n$ (needs **countable coproducts**)

“Polymorphic” List Transformations

Problem: Compute $\text{Nat}(\text{List}, \text{List}) = \text{all natural transformations } \alpha: \text{List} \Rightarrow \text{List}$

For example,

- $\text{rev} \in \text{Nat}(\text{List}, \text{List})$
- $(xs \mapsto []) \in \text{Nat}(\text{List}, \text{List})$
- $(\text{case } xs \text{ of } [] \mapsto []; [y \mid ys] \mapsto ys) \in \text{Nat}(\text{List}, \text{List})$

Natural transformations $P \Rightarrow Q$ are “polymorphic operators” from P to Q

Lemma (**PROVE IT**) :

$$\text{Nat}(P + R, Q) \cong \text{Nat}(P, Q) \times \text{Nat}(R, Q) \qquad \text{Nat}\left(\prod_{i \in I} P_i, Q\right) \cong \prod_{i \in I} \text{Nat}(P_i, Q)$$

Yoneda for Powers

- By previous lemma:

$$\text{Nat}(\text{List}, \text{List}) = \text{Nat}\left(\prod_{n \in \mathbb{N}} \text{Id}^n, \text{List}\right) \cong \prod_{n \in \mathbb{N}} \text{Nat}(\text{Id}^n, \text{List})$$

- But how to compute $\text{Nat}(\text{Id}^n, \text{List})$?

More generally, how to compute $\text{Nat}(\text{Id}^n, F)$ for $F: \mathbf{Set} \rightarrow \mathbf{Set}$?

- Take $\alpha: \text{Id}^n \Rightarrow F$. Then $\alpha_n: n^n \rightarrow F(n)$, so we can obtain $\alpha_n(\text{id}_n: n \rightarrow n) \in F(n)$
- Conversely: take $a \in F(n)$. Then $\alpha_X: X^n \rightarrow F(X)$, sending

$$(f: n \rightarrow X) \mapsto (F(f): F(n) \rightarrow F(X))(a)$$

is natural transformation (**PROVE IT**)

- These transitions form isomorphism (**PROVE IT**): $\text{Nat}(\text{Id}^n, F) \cong F(n)$

Natural List Transformers

Combine results of the previous slide with $F = \text{List}$:

$$\text{Nat}(\text{List}, \text{List}) \cong \prod_{n \in \mathbb{N}} \text{Nat}(\text{Id}^n, \text{List}) \cong 0^0 \times 1^1 \times 2^2 \times \dots$$

Action of $t \in \left([], [1], [k_1^2, k_2^2], [k_1^3, k_2^3, k_3^3], \dots \right)$:

$$[x_1, \dots, x_n] \in \text{List}(X) \quad \mapsto \quad [x_{k_1}, \dots, x_{k_n}] \in \text{List}(X)$$

Examples (Revisited):

- *rev*: $t = ([], [1], [2, 1], [3, 2, 1], \dots)$
- $(xs \mapsto [])$: $t = ([], [], [], \dots)$
- (*case* xs of $[] \mapsto []; [y \mid ys] \mapsto ys$): $t = ([], [], [2], [2, 3], \dots)$

Yoneda Theorem

Theorem: For any category $\mathcal{C}$, functor $F: \mathcal{C} \rightarrow \mathbf{Set}$ and object $A \in \text{Ob}(\mathcal{C})$:

$$\text{Nat}(\text{Hom}_{\mathcal{C}}(A, -), F) \cong F(A)$$

naturally in F and A (p. 49):

$$\begin{array}{ccccc} \text{Nat}(\text{Hom}(A, -), F) & \xrightarrow{\cong} & F(A) & & \\ \sigma \circ (-) \downarrow & \sigma: F \Rightarrow G & \downarrow \sigma_A & & \\ \text{Nat}(\text{Hom}(A, -), G) & \xrightarrow{\cong} & G(A) & & \end{array}$$

$$\begin{array}{ccccc} \text{Nat}(\text{Hom}(A, -), F) & \xrightarrow{\cong} & F(A) & & \\ \alpha \mapsto \alpha \circ \text{Hom}(f, -) \downarrow & f: A \rightarrow B & \downarrow F(f) & & \\ \text{Nat}(\text{Hom}(B, -), F) & \xrightarrow{\cong} & F(B) & & \end{array}$$

Under: $(\alpha: \text{Hom}_{\mathcal{C}}(A, -) \Rightarrow F) \mapsto \alpha_A(id_A)$

$(f: A \rightarrow B \mapsto F(f)(a))_{B \in \text{Ob}(\mathcal{C})} \leftarrow a \in F(A)$

More on Naturality

The bijection $\text{Nat}(\text{Hom}_{\mathcal{C}}(A, -), F) \cong F(A)$ is **jointly** natural in $F \in [\mathcal{C}, \mathbf{Set}]$ and $A \in \mathcal{C}$:

$$\begin{array}{ccc} & (F, A) \mapsto \text{Nat}(\text{Hom}(A, -), F) & \\ & \downarrow \cong & \\ [\mathcal{C}, \mathbf{Set}] \times \mathcal{C} & & \mathbf{Set} \\ & (F, A) \mapsto F(A) & \end{array}$$

Both sides define functors $[\mathcal{C}, \mathbf{Set}] \times \mathcal{C} \rightarrow \mathbf{Set}$; the Yoneda bijection is a natural isomorphism between them

Bottom **evaluation functor** manifests Cartesian closure of **Cat**!

Yoneda Embedding

Yoneda Embedding: $y: \mathcal{C}^{\text{op}} \rightarrow [\mathcal{C}, \mathbf{Set}]$ by $y(A) = \text{Hom}_{\mathcal{C}}(A, -)$

Corollary I: y is faithful: if $y(f) = y(g)$ then $f = g$

Proof: Isomorphism

$$\text{Nat}(\text{Hom}(A, -), \text{Hom}(B, -)) \cong \text{Hom}(B, A) \quad (*)$$

sends $y(f)$ and $y(g)$ to $\text{Hom}(f, A)(id_A) = f$ and $\text{Hom}(g, A)(id_A) = g$

Corollary II: y is full: every $\alpha: \text{Hom}(A, -) \Rightarrow \text{Hom}(B, -)$ equals $y(f)$ for some $f: B \rightarrow A$

Proof: Isomorphism $(*)$ sends α to $\alpha_A(id_A)$ and $y(f)$ to f where $f = \alpha_A(id_A)$

Dually (**co-Yoneda**): take $\mathcal{C} := \mathcal{C}^{\text{op}} \rightsquigarrow y': \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ by $y'(A) = \text{Hom}_{\mathcal{C}}(-, A)$

Yoneda for Posets

Let $(P, \leq)$ be poset category. Then

- $\text{Hom}(-, a) = \downarrow a = \{b \mid b \leq a\} : P^{\text{op}} \rightarrow \mathbf{Set}$ (**principal ideals**)
- $\alpha : \text{Hom}(-, a) \Rightarrow \text{Hom}(-, b)$ is wittiness of $\downarrow a \subseteq \downarrow b$

Co-Yoneda Embedding: $y' : P \rightarrow [P^{\text{op}}, \mathbf{Set}]$, $a \mapsto \downarrow a$, **factors through downset completion:**

$$P \xrightarrow{a \mapsto \downarrow a} [P^{\text{op}}, 2] = \{D \subseteq P \mid D \text{ is down-closed}\} \hookrightarrow [P^{\text{op}}, \mathbf{Set}]$$

Faithfulness and Fullness: $\downarrow a = \downarrow b \Rightarrow a = b$, $\downarrow a \subseteq \downarrow b \iff a \leq b$

Upshot: $[P^{\text{op}}, 2]$ = free completion of P to complete lattice (all joins freely added); For general categories: $[C^{\text{op}}, \mathbf{Set}]$ = free completion of C under all colimits

Adjunctions & Free Objects

Rounding Up and Down

- View $\mathbb{Z}, \mathbb{R}$ as poset categories; inclusion $\iota: \mathbb{Z} \hookrightarrow \mathbb{R}$ monotone $\Rightarrow$ functor
- For $n \in \mathbb{Z}, r \in \mathbb{R}$:

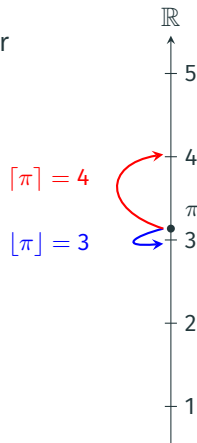
$$r \leq \iota(n) \Leftrightarrow \lceil r \rceil \leq n \qquad \iota(n) \leq r \Leftrightarrow n \leq \lfloor r \rfloor$$

- Categorically:

$$\text{Hom}(r, \iota(n)) \cong \text{Hom}(\lceil r \rceil, n) \quad - \quad \lceil - \rceil \text{ is left adjoint to } \iota$$

$$\text{Hom}(\iota(n), r) \cong \text{Hom}(n, \lfloor r \rfloor) \quad - \quad \lfloor - \rfloor \text{ is right adjoint to } \iota$$

- **Adjunctions** are pairs (left adjoint, right adjoint)
- Adjunctions between order categories are also called **Galois connections**



Adjunctions

Functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{C}$ form **adjunction** $F \dashv G$ (**F** left adjoint, **G** right adjoint) if there is bijection

$$\varphi_{X,Y}: \text{Hom}_{\mathcal{D}}(F(X), Y) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(X, G(Y))$$

natural in $X \in \mathcal{C}$ and $Y \in \mathcal{D}$:

$$\begin{array}{ccccc} \text{Hom}(F(X), Y) & \xrightarrow{\varphi_{X,Y}} & \text{Hom}(X, G(Y)) & & \\ - \circ F(f) \downarrow & f: X' \rightarrow X & \downarrow - \circ f & & \\ \text{Hom}(F(X'), Y) & \xrightarrow{\varphi_{X',Y}} & \text{Hom}(X', G(Y)) & & \end{array}$$

$$\begin{array}{ccccc} \text{Hom}(F(X), Y) & \xrightarrow{\varphi_{X,Y}} & \text{Hom}(X, G(Y)) & & \\ g \circ - \downarrow & g: Y \rightarrow Y' & \downarrow G(g) \circ - & & \\ \text{Hom}(F(X), Y') & \xrightarrow{\varphi_{X,Y'}} & \text{Hom}(X, G(Y')) & & \end{array}$$

Mnemonics: F for **F**ree, G for **G**etful

Examples of Adjunctions

- **Exponentials:** currying/uncurrying bijection

$$\mathrm{Hom}(Z \times X, Y) \cong \mathrm{Hom}(Z, Y^X)$$

witnesses $\boxed{(- \times X) \dashv (-)^X}$ in any CCC

- **Coproducts:**

$$\mathrm{Hom}_{\mathcal{C}}(A + B, C) \cong \mathrm{Hom}_{\mathcal{C}}(A, C) \times \mathrm{Hom}_{\mathcal{C}}(B, C) = \mathrm{Hom}_{\mathcal{C} \times \mathcal{C}}((A, B), (C, C))$$

witnesses $\boxed{+ \dashv \Delta}$ where $\Delta: \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$, $\Delta(C) = (C, C)$

- **Lists:** $\boxed{\mathrm{List} \dashv G}$ for forgetful functor $G: \mathbf{Mon} \rightarrow \mathbf{Set}$, because

$$\mathrm{Hom}_{\mathbf{Mon}}(\mathrm{List}(X), M) \cong \mathrm{Hom}_{\mathbf{Set}}(X, G(M))$$

List as Left Adjoint

Check $\text{Hom}_{\mathbf{Mon}}(\text{List}(X), M) \cong \text{Hom}_{\mathbf{Set}}(X, G(M))$:

- Monoid morphism $f: \text{List}(X) \rightarrow M$ (for monoid $(M, e, \cdot)$):

$$f([]) = e$$

$$f([x_0, \dots, x_n]) = f([x_0]) \cdot \dots \cdot f([x_n])$$

- Thus, f is determined by its action on $x_0, \dots, x_n \in X$
- Conversely, if $f: X \rightarrow M$, we can **extend** it to entire $\text{List}(X)$:

$$f^*([]) = e$$

$$f^*([x_0, \dots, x_n]) = f(x_0) \cdot \dots \cdot f(x_n)$$

- ... and it is a monoid morphism, because

$$f^*(xs ++ ys) = f^*(xs) \cdot f^*(ys)$$

Unit and Counit

Considering $F \dashv G$ with $\varphi_{X,Y}: \text{Hom}_{\mathcal{D}}(F(X), Y) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(X, G(Y))$:

- apply $\varphi_{X,F(X)}$ to $\text{id}_{F(X)}$, get **unit** $\eta_X = \varphi_{X,F(X)}(\text{id}_{F(X)}): X \rightarrow GF(X)$
- dually, apply $\varphi_{G(Y),Y}^{-1}$ to $\text{id}_{G(Y)}$ to get **counit**
- Unit and counit are both natural transformations (**PROVE IT**)

Definition: **Equivalence** of categories is such adjunction $F \dashv G$, whose unit and counit are natural isomorphisms

In Summary:

Isomorphism < Equivalence < Adjunction

Units and Counits: Examples

- **Exponentials:** $(- \times X) \dashv (-)^X$ in a CCC,
 - Unit $\eta_Z: Z \rightarrow (Z \times X)^X$, $\eta = \text{curry}(id)$
 - Counit $\varepsilon_Y: Y^X \times X \rightarrow Y$, $\varepsilon = ev$
- **Lists:** $\text{List} \dashv G$,
 - Unit $\eta_X: X \rightarrow G(\text{List}(X))$, $\eta(x) = [x]$
 - Counit $\varepsilon_M: \text{List}(G(M)) \rightarrow M$, $\varepsilon([m_0, \dots, m_n]) = m_0 \cdot \dots \cdot m_n$
- **Coproducts:** $+ \dashv \Delta$, where $\Delta(C) = (C, C)$,
 - Unit $\eta_{(A,B)} = (inl: A \rightarrow A + B, inr: B \rightarrow A + B)$
 - Counit $\varepsilon_C: C + C \rightarrow C$, $\varepsilon = [id, id]$

Adjunctions & Free Objects

Free Objects

Lists as Free Monoids

Fix set X

- **Object:** $\text{List}(X)$ in **Mon**
- **Arrows:** Unit map $\text{sngl}_X: X \rightarrow \text{List}(X)$ in **Set**, $x \mapsto [x]$
- **Universal property:** For any monoid M and $f: X \rightarrow M$ in **Set**, there exists a **unique** monoid morphism $f^*: \text{List}(X) \rightarrow M$ such that:

$$\begin{array}{ccc} \text{List}(X) & \xrightarrow{\quad f^* \quad} & M \\ \uparrow \text{sngl}_X & \nearrow f & \\ X & & \end{array}$$

Note that f is from **Set**, but f^* is from **Mon**; $G: \mathbf{Mon} \rightarrow \mathbf{Set}$ is the forgetful functor

Lists as Free Monoids

Fix set X

- **Object:** $\text{List}(X)$ in **Mon**
- **Arrows:** Unit map $\text{sngl}_X: X \rightarrow \text{List}(X)$ in **Set**, $x \mapsto [x]$
- **Universal property:** For any monoid M and $f: X \rightarrow M$ in **Set**, there exists a **unique** monoid morphism $f^*: \text{List}(X) \rightarrow M$ such that:

$$\begin{array}{ccc} G(\text{List}(X)) & \xrightarrow{G(f^*)} & G(M) \\ \uparrow \text{sngl}_X & \nearrow f & \\ X & & \end{array}$$

Note that f is from **Set**, but f^* is from **Mon**; $G: \mathbf{Mon} \rightarrow \mathbf{Set}$ is the forgetful functor

Free Objects

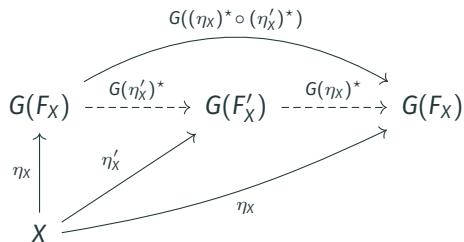
Fix $G: \mathcal{D} \rightarrow \mathcal{C}$ and $X \in \text{Ob}(\mathcal{C})$

- **Object:** F_X in $\mathcal{D}$
- **Arrows:** morphism $\eta_X: X \rightarrow G(F_X)$ in $\mathcal{C}$
- **Universal property:** for any $Y \in \mathcal{D}$ and $f: X \rightarrow G(Y)$ in $\mathcal{C}$, there exists unique **extension** $f^*: F_X \rightarrow Y$ in $\mathcal{D}$ such that:

A commutative triangle diagram illustrating the universal property of free objects. The vertices are X at the bottom, $G(F_X)$ at the top-left, and $G(Y)$ at the top-right. A vertical arrow labeled η_X points from X to $G(F_X)$. A diagonal arrow labeled f points from X to $G(Y)$. A dashed horizontal arrow labeled $G(f^*)$ points from $G(F_X)$ to $G(Y)$.

Free Objects are Up-to-Isomorphism

If (F_X, η_X) and (F'_X, η'_X) are both free objects on $X \in \mathcal{C}$, apply universal properties:



Both $(\eta_X)^* \circ (\eta'_X)^*$ and id_{F_X} extend η_X to $G(F_X)$, so by uniqueness: $(\eta_X)^* \circ (\eta'_X)^* = id_{F_X}$; other direction analogous, hence $F_X \cong F'_X$

Free Object Functor

If all free objects F_X exist then F extends to functor and η to natural transformation:

$$\begin{array}{ccccc} & & G(\eta_Y \circ f \circ g)^* & & \\ & \nearrow & & \searrow & \\ G(F_X) & \overset{G(\eta_Y \circ f)^*}{\dashrightarrow} & G(F_Y) & \overset{G(\eta_Z \circ g)^*}{\dashrightarrow} & G(F_Z) \\ \eta_X \uparrow & & \eta_Y \uparrow & & \eta_Z \uparrow \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z \end{array}$$

$$\begin{array}{ccc} G(F_X) & \overset{id}{\dashrightarrow} & G(F_X) \\ \eta_X \uparrow & \nearrow \eta_X & \\ X & & \end{array}$$

So, we take $F(X) = F_X$ and $F(f) = (\eta_Y \circ f)^*$

Co-Free Objects

- Recall our definition of exponentials:

$$\begin{array}{ccc}
 Z \times X & \xrightarrow{\text{curry}_{X,Y,Z}(f) \times id_X} & Y^X \times X \\
 & \searrow f & \downarrow ev_{X,Y} \\
 & & Y
 \end{array}$$

- These can be viewed as **co-free objects** w.r.t. $F = (-) \times X$
- Dually, $FZ = Z \times X$ is free object w.r.t. $G = (-)^X$ (**PROVE IT**):

$$\begin{array}{ccc}
 (Z \times X)^X & \xrightarrow{(\text{uncurry}_{X,Y,Z})^X} & Y^X \\
 \uparrow \text{curry}_{X,Z \times X,Z}(id_{Z \times X}) & \nearrow f & \\
 Z & &
 \end{array}$$

Adjunctions via Free Objects

- Recall that we obtained $uncurry = curry^{-1}$ and adjunction

$$\text{Hom}(Z \times X, Y) \cong \text{Hom}(Z, Y^X) :$$

from $(Y^X, ev_{X,Y})$ as co-free object

- It is a general principle!

Proposition: Given $G: \mathcal{D} \rightarrow \mathcal{C}$, $F \dashv G$ exists iff all free objects $F(X)$ w.r.t. G exist

- Adjunction $\Rightarrow$ Free objects:** given $\varphi: \text{Hom}(F(X), Y) \cong \text{Hom}(X, G(Y))$, set $\eta_X = \varphi(id_{F(X)})$; for $f: X \rightarrow G(Y)$ define $f^* = \varphi^{-1}(f): F(X) \rightarrow Y$
- Free objects $\Rightarrow$ Adjunction:** given $(F(X), \eta_X)$, define

$$\varphi(h) = G(h) \circ \eta_X \quad \varphi^{-1}(f) = \text{unique } f^* \text{ with } G(f^*) \circ \eta_X = f$$

Corollary: Left adjoints are unique up to isomorphism (because so are free objects)

Adjunctions: Summary

We can define adjunctions

1. through natural isomorphism of homs
2. through free objects
3. through co-free objects
4. through units and counits (triangle identities omitted)

Note: 1. and 2. allow us to deal with particular (co-)free object even when adjunction need not exist globally!

Conclusions

Topics covered (or so I hope):

- Sets, functions, naturals, (co-)products in set theory
- Categories, functors, isomorphisms, products, coproducts
- Natural transformations, Yoneda lemma, CCCs, exponentials
- Adjunctions, free objects, units, counits

Further topics:

- Monads and coalgebras
- Limits and colimits
- Topos theory
- Enriched categories, monoidal categories, 2-categories and higher categories thereof
- ...

Thank You for Your Attention!